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PROBABILISTIC ACTION PREDICTION OF HUMAN DRIVERS USING DYNAMIC BAYESIAN NETWORKS

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ABSTRACT

Autonomous vehicles (AVs) are expected to operate for many years in mixed traffic with human-driven vehicles (HDVs), where estimating human maneuvers is important for safe interaction. This paper presents a probabilistic framework for short-horizon human driver maneuver prediction using Dynamic Bayesian Networks (DBNs). The model is trained on the highD naturalistic highway dataset using window-based observations from vehicle motion and surrounding traffic interactions. The DBN uses a hierarchical latent structure, where action states are conditioned on broader driving regimes, allowing maneuver type and interaction context to be modeled together.

On the test set, the model gives a promising performance, with 79.4% exact next-step accuracy, 67.6% macro precision, 71.5% macro F1-score, and an 81.3% hit rate over a 2 s prediction horizon. The average CPU latency for online belief update and next-maneuver prediction is 0.61 ms per vehicle, far below the 400 ms input interval. The results show that DBNs can support fast and interpretable short-horizon maneuver prediction.

Keywords: Human Driver Behavior Modeling, Autonomous Driving, Human-Driven Vehicles, Dynamic Bayesian Networks, Maneuver Prediction, Probabilistic Modeling, Mixed Traffic Environments

1. INTRODUCTION

Recent progress in artificial intelligence (AI), sensing, computing, and communication has accelerated the development of AVs. However, safe operation still depends on how well AVs can handle mixed traffic with HDVs, since fully autonomous traffic systems do not yet exist.

In mixed traffic, an AV cannot directly observe the intentions of nearby HDVs. Instead, human driver behavior must be inferred from observable cues such as vehicle motion, lane changes, relative positions, and interactions with other nearby vehicles [1]. Human driver behavior prediction is therefore an important problem in autonomous driving. Estimating likely future maneuvers can support downstream tasks such as risk assessment, trajectory prediction, and motion planning [1–3].

Existing behavior prediction methods include rule-based, probabilistic, and learning-based approaches [2, 4, 5]. Rule-based models are simple and interpretable but often too rigid to capture the variability of real human behavior. Learning-based methods can learn complex patterns from large datasets, but many are difficult to interpret and are black boxes. Probabilistic models provide a useful middle ground because they can represent uncertainty, capture temporal dependencies, and remain interpretable.

This paper focuses on probabilistic modeling of high-level human driving behavior in highway traffic, with the goal of predicting likely future maneuvers of surrounding drivers while explicitly representing uncertainty. In the proposed framework, future maneuvers are represented as discrete action states, and a DBN models the temporal relationship between trajectory-based observations and hidden behavioral states. Driver intent is represented through latent variables and inferred probabilistically from observed vehicle motion and traffic context [6]. The model is trained and evaluated on the highD dataset [7] using features that describe vehicle motion, surrounding traffic context, and interaction behavior.

The study evaluates whether this framework can provide accurate, interpretable, and computationally efficient short-horizon predictions of future human driving maneuvers.

The main contributions of this paper are threefold: (1) a structured DBN framework for probabilistic short-horizon maneuver prediction with interpretable latent states, (2) a window-based observation design that combines ego motion with surrounding traffic interaction features, and (3) an evaluation on the highD dataset covering predictive accuracy, short-horizon hit rate, and online inference latency.

2. RELATED WORK

Human driver behavior and maneuver prediction methods are commonly grouped into rule-based, probabilistic, and learning-based approaches, which differ in how they represent driver decisions, uncertainty, and temporal dependencies.

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2.1. Rule-Based Driver Models

Rule-based models describe driving behavior through explicit mathematical equations or logical rules. Typical examples include car-following and lane-change models such as the Intelligent Driver Model (IDM), Gipps' model and the Minimizing Overall Braking Induced by Lane Changes (MOBIL) model, which relate observable traffic variables to longitudinal or lateral driving actions [8–11].

Their main strengths are interpretability, low computational cost, and clear physical meaning of parameters. However, they usually rely on hand-crafted assumptions and often represent driver behavior in a simplified or mostly deterministic way. This limits their ability to capture uncertainty and variability across drivers in real traffic.

2.2. Probabilistic Temporal Models

Probabilistic models address uncertainty by describing driver behavior as a stochastic process. Instead of predicting a single fixed response, they represent multiple possible behaviors and quantify uncertainty over future actions [12]. This is useful for human-driver prediction because the same observed traffic situation can lead to different driver responses.

Among probabilistic methods, Hidden Markov Models (HMMs) are a common starting point for modeling time-varying driving behavior. They represent behavior as a sequence of latent states that generate observable signals [13]. In driver modeling, these hidden states can correspond to maneuver-related modes such as lane keeping, braking, or overtaking [14]. HMMs are attractive because of their simple structure and efficient inference. However, they represent the hidden state with a single variable, which limits how much structure can be encoded in the model.

DBNs extend this temporal modeling idea by representing the state of a system with several variables across time [6, 15]. This makes it possible to encode temporal dependencies between hidden variables, observations, and contextual factors in a structured probabilistic model. In driver-behavior research, Bayesian-network-based models have been used for maneuver prediction, lane-change reasoning, and safety-related inference [16–18].

Li et al. proposed a DBN-based lane-change maneuver prediction framework for highway driving using historical vehicle states, road structure, and traffic interaction features. Their model achieved an F1-score of 80% for lane-change recognition and an advance prediction time of 3.75 s [16]. Talluri et al. used a DBN to combine lateral evidence with safety assessment for cut-in and lane-change reasoning [18]. Mlynarczyk et al. focused on an affective driver-behavior module, where Bayesian networks estimate mental states such as mental load and fatigue from physiological and demographic variables [17]. In contrast, the present work learns a compact DBN directly from naturalistic highD trajectory windows, using a factored latent structure with style and action variables. The learned joint latent states are mapped to semantic maneuver labels only after training, allowing short-horizon maneuver prediction while keeping the latent behavior representation interpretable.

Other probabilistic approaches, such as Gaussian Mixture Models (GMMs), Gaussian Mixture Regression (GMR), and Gaussian Process Regression (GPR), are also used in driving-behavior and trajectory modeling [19–21]. These methods are useful for continuous quantities such as acceleration or trajectory evolution, but the present paper focuses on discrete short-horizon maneuver prediction with a structured temporal model.

2.3. Learning-Based Prediction Models

Learning-based methods form another major family of behavior prediction models. Classical machine learning methods such as Support Vector Machine (SVM), decision trees, and random forests were early data-driven approaches for intent recognition and maneuver classification [22]. Recent data-driven models have reported high performance for lane-change-specific tasks on highway trajectory datasets. For example, Xue et al. reported lane-change decision accuracy increasing from 97.02% to 98.20% when traffic context was included, while Geng et al. reported 98.82% overall accuracy for lane-changing behavior prediction on highD [23, 24]. These results are strong, but they mainly address the specific task of lane-change prediction. They do not directly solve the broader problem of learning latent human-driving regimes and predicting high level maneuver types in one compact probabilistic model.

Deep learning has become a major direction in trajectory and maneuver prediction. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were widely used to model temporal dependencies in driving sequences, while Graph Neural Networks (GNNs) were later introduced to better represent interactions between multiple traffic participants [25, 26]. More recently, Transformer-based models have shown strong performance by capturing long-range dependencies and multi-agent interactions through attention mechanisms [27–30]. Generative approaches such as Conditional Variational Autoencoder (CVAE) and diffusion models further support multimodal prediction by generating multiple plausible future trajectories under the same observed context [26, 31–34].

Although these methods are powerful, especially for continuous trajectory forecasting and lane-change-specific classification, most of them often operate as highly data-driven black-box models. For applications where uncertainty representation, temporal structure, and behavioral interpretability are important, this can be a limitation. However, the goal of this paper is different. The focus is not full trajectory generation or lane-change detection, but short-horizon prediction of discrete human-driver maneuvers with an interpretable model.

2.4. Positioning of This Work

The review above shows a trade-off between simplicity, predictive power, and interpretability. Rule-based models are transparent but limited in uncertainty representation. They depend strongly on hand-crafted assumptions. Learning-based models can achieve high prediction accuracy, especially for supervised lane-change or trajectory forecasting tasks, but they often require large labeled datasets and careful tuning. Probabilistic temporal models provide a useful option when the aim is to model uncertainty and update predictions over time.

This paper uses a DBN because the target problem is short-horizon maneuver prediction from naturalistic highway trajectories, where semantic maneuver labels are not directly provided by the dataset. The proposed model learns latent behavior states from window-level motion and interaction features, then maps the learned latent states to semantic maneuver labels after training. This allows the model to perform online probabilistic prediction while keeping the learned behavior representation compact and inspectable. The work therefore focuses on structured human-driver maneuver prediction for HDVs, rather than full trajectory generation or downstream planning.

3. METHODOLOGY

This section describes the methodology used to construct and evaluate the proposed DBN-based maneuver prediction framework. It first presents the overall workflow and the selected highD data subset. It then explains the preprocessing steps, the sliding-window observation design, the DBN formulation, and the training and online prediction procedure. The full processing pipeline is summarized in Fig. 1.

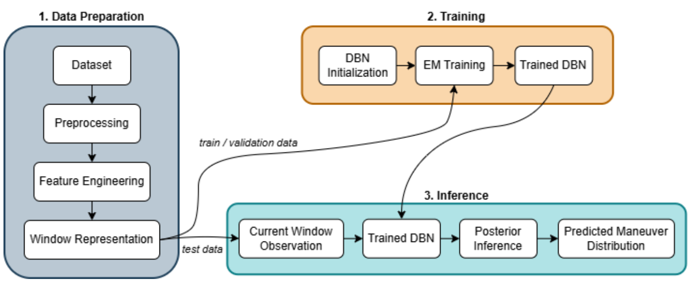


FIGURE 1: Overview of the proposed framework

3.1. Framework Overview

The proposed framework consists of three main stages: preprocessing of raw trajectory data, probabilistic temporal modeling, and causal online prediction. First, frame-level trajectories from highD are converted into a consistent representation that is independent of the original driving direction. Next, ego-motion, lane-related, interaction, and risk-related features are extracted and aggregated over short overlapping windows. Each window then forms one observation for the DBN, which models temporal evolution through two latent variables: a style state and an action state. After training, the model is used for online filtering and one-step-ahead maneuver prediction.

3.2. Dataset and Preprocessing

The method is evaluated on the highD dataset [7], which contains naturalistic highway trajectories recorded from drone videos on German highways. For each vehicle, the dataset provides frame-level position, velocity, acceleration, lane information, neighboring-vehicle identifiers, and traffic-safety indicators such as time headway (THW) and time to collision (TTC).

Since the recordings contain traffic moving in opposite roadway directions, the raw kinematic signals do not follow a consistent sign convention across all vehicles. A vehicle-centric normalization step is therefore applied so that the same maneuver is represented with the same sign convention regardless of the original travel direction. This produces a consistent frame-level representation for later feature extraction and windowization.

3.3. Data Analysis for Feature Design

Before defining the final observation vector, an exploratory analysis of the dataset was performed. The goal of this step was to understand which signals carry useful information for maneuver prediction and how they should be represented in the observation vector. The analysis focused on three aspects: frame-level kinematics, trajectory-level variability, and lane-change sparsity.

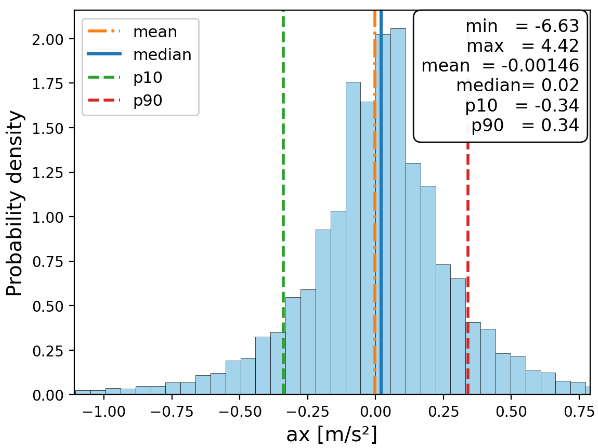
Fig. 2 shows the global distributions of the main kinematic variables. The longitudinal velocity v_x covers a broad range, which reflects different traffic speeds and vehicle classes. The longitudinal acceleration a_x is centered close to zero, since many vehicles drive with almost steady speed, but it also has visible positive and negative tails. These tails are important because they describe acceleration and braking behavior, which are directly related to maneuver changes and car-following reactions.

In contrast, the lateral variables v_y and a_y are strongly concentrated around zero. This is expected because most vehicles stay inside their lane for most frames. However, both variables still show non-zero tails. These tails are useful because they contain information about lateral adjustments and lane-change motion. Therefore, even though lateral motion is sparse compared with longitudinal motion, it should not be removed from the observation vector.

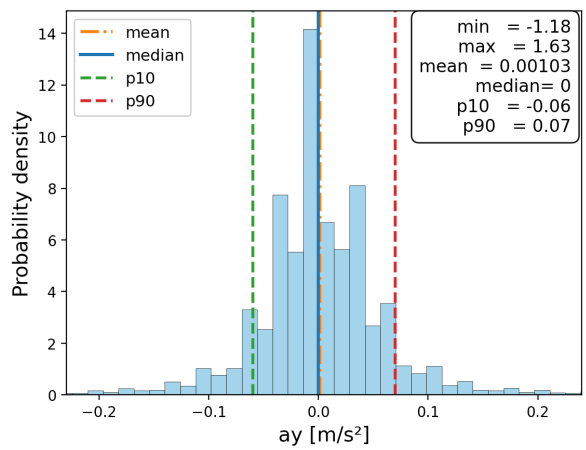
Trajectory-level variability was also analyzed to capture how much each vehicle changes its motion over time. For each vehicle trajectory, the standard deviation of v_x, a_x, v_y, and a_y was computed. This gives one variability value per trajectory for each signal. As shown in Fig. 4, the longitudinal variables show larger variability than the lateral variables. This means that speed adaptation and acceleration changes occur more often and with larger magnitude than lateral movement. At the same time, some trajectories show stronger lateral variability, which usually corresponds to lane changes or larger lateral corrections.

The lane-change analysis is also shown in Fig. 3. Lane keeping dominates the data, while left and right lane changes form only a small fraction of the frames. The same imbalance appears at trajectory level, where most vehicles perform no lane change and only a small number of vehicles perform one or more lane changes. This imbalance is important for feature design. A single lane-change indicator alone is too sparse to provide a strong learning signal. For this reason, the final observation vector also includes lateral velocity, lateral acceleration, lane-boundary distances, neighboring-vehicle context, and temporally aggregated lane-change evidence. These additional features help the model detect the build-up and context of a maneuver, not only the exact frame where the lane ID changes.

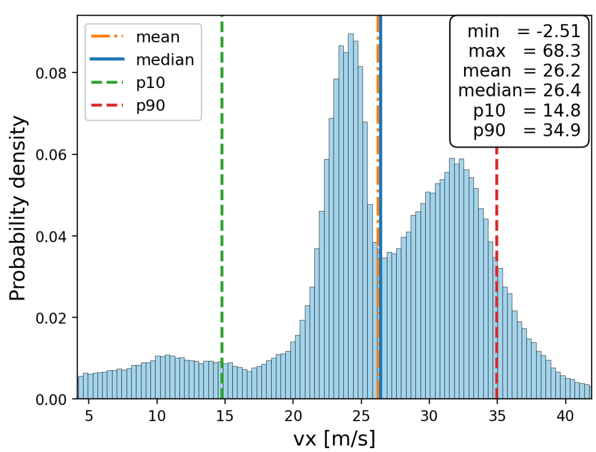
Overall, the analysis led to three design choices. First, both longitudinal and lateral kinematic signals are included because they describe complementary parts of driver behavior. Second, the model uses temporal window summaries instead of only single-frame values, since variability and short-term trends contain useful maneuver information. Third, sparse lane-change evidence is combined with motion, lane geometry, interaction, and risk-related features so that the model can learn the driving context around rare maneuvers.



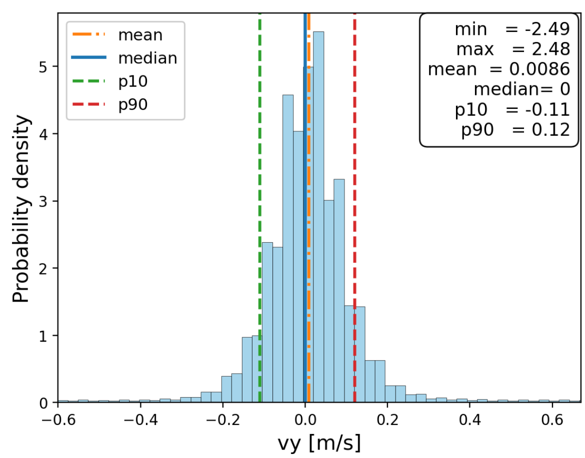
(a) Longitudinal acceleration



(b) Lateral acceleration



(c) Longitudinal velocity



(d) Lateral velocity

FIGURE 2: Distributions of the main frame-level kinematic variables.

3.4. Window-Based Observation Design

The observation vector combines ego-motion, lane-related, interaction, and risk-related information. It includes kinematic quantities such as longitudinal and lateral velocity and acceleration, lane-boundary distances, lane-change evidence, front-vehicle safety indicators, and selected relative features describing nearby traffic participants. These variables are designed to represent both the state of the ego vehicle and the surrounding traffic context relevant for maneuver prediction.

Rather than using individual frames directly, the model operates on short temporal windows. Each trajectory is converted into a sequence of overlapping windows with length $W = 50$ frames and stride $r = 10$ frames. Since the highD dataset is recorded at 25 Hz, each window covers 2.0 s of history and consecutive windows start every 0.4 s. Each window is summarized into one observation vector using simple temporal aggregation functions, which provide a compact representation of recent motion and interaction history while keeping the observation space manageable. The aggregation includes last values, window minima, temporal slopes, sign fractions, and binary event indicators, depending on the feature type. The resulting windowization process is illustrated in Fig. 5.

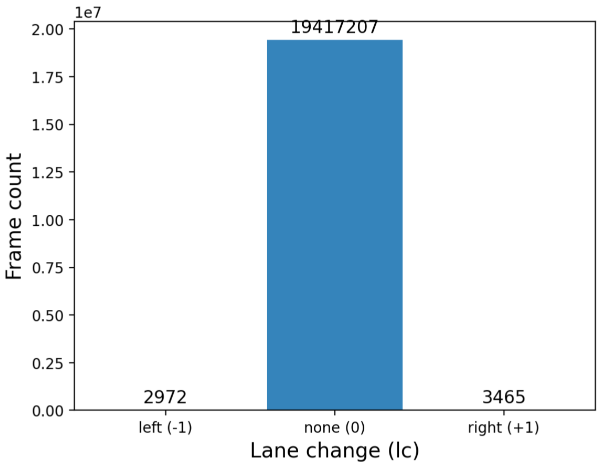
3.5. DBN Formulation

The proposed DBN uses two discrete latent variables at each time step: a style state S_t and an action state A_t. The style state captures a broader driving regime, while the action state represents the current maneuver. The observation vector O_t corresponds to the aggregated window-level features described above. The graphical structure in Fig. 6 shows the observation as a single node for compactness. In the implementation, this observation node is split into an action-related block and a style-related block.

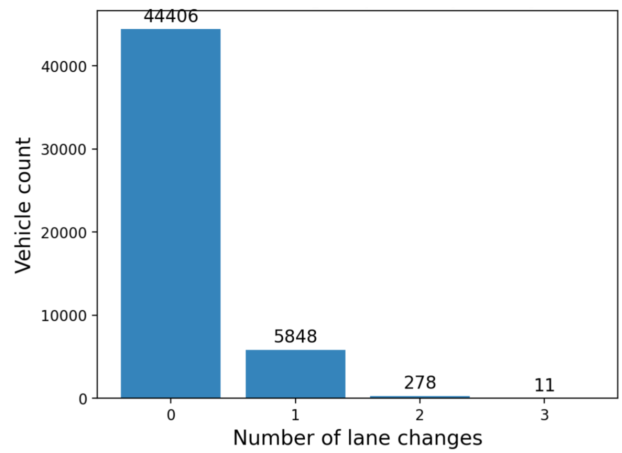
The temporal dependencies are factorized such that the style state evolves according to its previous value, while the action state depends on both the previous action and the current style. This structure reflects the assumption that the broader driving regime changes more slowly, whereas the short-term maneuver depends on both recent behavior and current traffic context.

The observation sequence for one vehicle is denoted by $O_{1:T} = (O_1, \ldots, O_T)$, where each O_t is one window-level observation. In the emission model used in this paper, each observation is split into two feature blocks,

$$O_t = \left(O_t^{\text{act}}, O_t^{\text{sty}}\right). \tag{1}$$



(a) Raw lane-change indicator at frame level



(b) Number of lane changes per trajectory

FIGURE 3: Lane-change sparsity analysis.

The action-related block O_t^{act} contains ego-motion, lane-change, and lane-geometry features that describe the physical maneuver. The style-related block O_t^{sty} contains interaction, risk, and surrounding-traffic features that describe the driving context.

The joint distribution of the proposed DBN for a sequence of length T is written as

$$\begin{aligned} &p(S_{1:T}, A_{1:T}, O_{1:T}^{\text{act}}, O_{1:T}^{\text{sty}}) \\ &= p(S_1)p(A_1 \mid S_1)p(O_1^{\text{act}} \mid A_1)p(O_1^{\text{sty}} \mid S_1) \\ &\quad \times \prod_{t=2}^{T} p(S_t \mid S_{t-1})p(A_t \mid A_{t-1}, S_t) \\ &\quad \times p(O_t^{\text{act}} \mid A_t)p(O_t^{\text{sty}} \mid S_t). \end{aligned} \tag{2}$$

This factorization means that the action-related feature block is conditioned on the action state, while the style-related feature block is conditioned on the style state. The action state is still influenced by the current style state through the transition model $p(A_t \mid A_{t-1}, S_t)$.

The emission likelihood is therefore written as

$$p(O_t \mid S_t = s, A_t = a) = p(O_t^{\text{act}} \mid A_t = a)p(O_t^{\text{sty}} \mid S_t = s). \tag{3}$$

For each block $g \in \{\text{sty}, \text{act}\}$, continuous features are modeled with Gaussian distributions with diagonal covariance and binary features with Bernoulli distributions:

$$\begin{aligned} p(O_t^g \mid z_g) = &\prod_{d \in \mathcal{C}_g} \mathcal{N}\left(o_{t,d}; \mu_{z_g,d}, \sigma^2_{z_g,d}\right) \\ &\times \prod_{b \in \mathcal{B}_g} \text{Bern}\left(o_{t,b}; \rho_{z_g,b}\right), \end{aligned} \tag{4}$$

where $z_{\text{sty}} = s$ and $z_{\text{act}} = a$. The sets $\mathcal{C}_g$ and $\mathcal{B}_g$ denote the continuous and binary features in block g, respectively. The parameters $\mu_{z_g,d}$ and $\sigma^2_{z_g,d}$ define the Gaussian emission for continuous feature d, while $\rho_{z_g,b}$ defines the Bernoulli probability for binary feature b. This diagonal-covariance choice is a tractable emission-model approximation; it does not imply that the raw motion features are independent in the data.

In the experiments reported in this paper, the DBN uses two style states and four action states, denoted as S2-A4 (where S = style states and A = action states, giving $2 \times 4 = 8$ possible joint latent states). Each joint state is later mapped to a semantic maneuver label for interpretation and predictive evaluation.

3.6. Training and Online Prediction

Model parameters are estimated from the training data using the expectation-maximization (EM) algorithm. Continuous features are normalized using statistics computed from the training split only, and model initialization includes initial-state probabilities, transition probabilities, and emission parameters derived from the training data. The training procedure learns both the temporal dynamics of the latent states and the emission model linking latent states to window-level features.

After training, the model is used in a causal online setting. At time step t, the filtered belief over the latent state pair is updated using the observations available up to that time, and the transition model is then used to obtain the one-step-ahead predictive distribution over future maneuvers. The causal prediction process is illustrated in Fig. 7.

After filtering, the one-step-ahead predictive distribution over the next joint latent state is obtained as

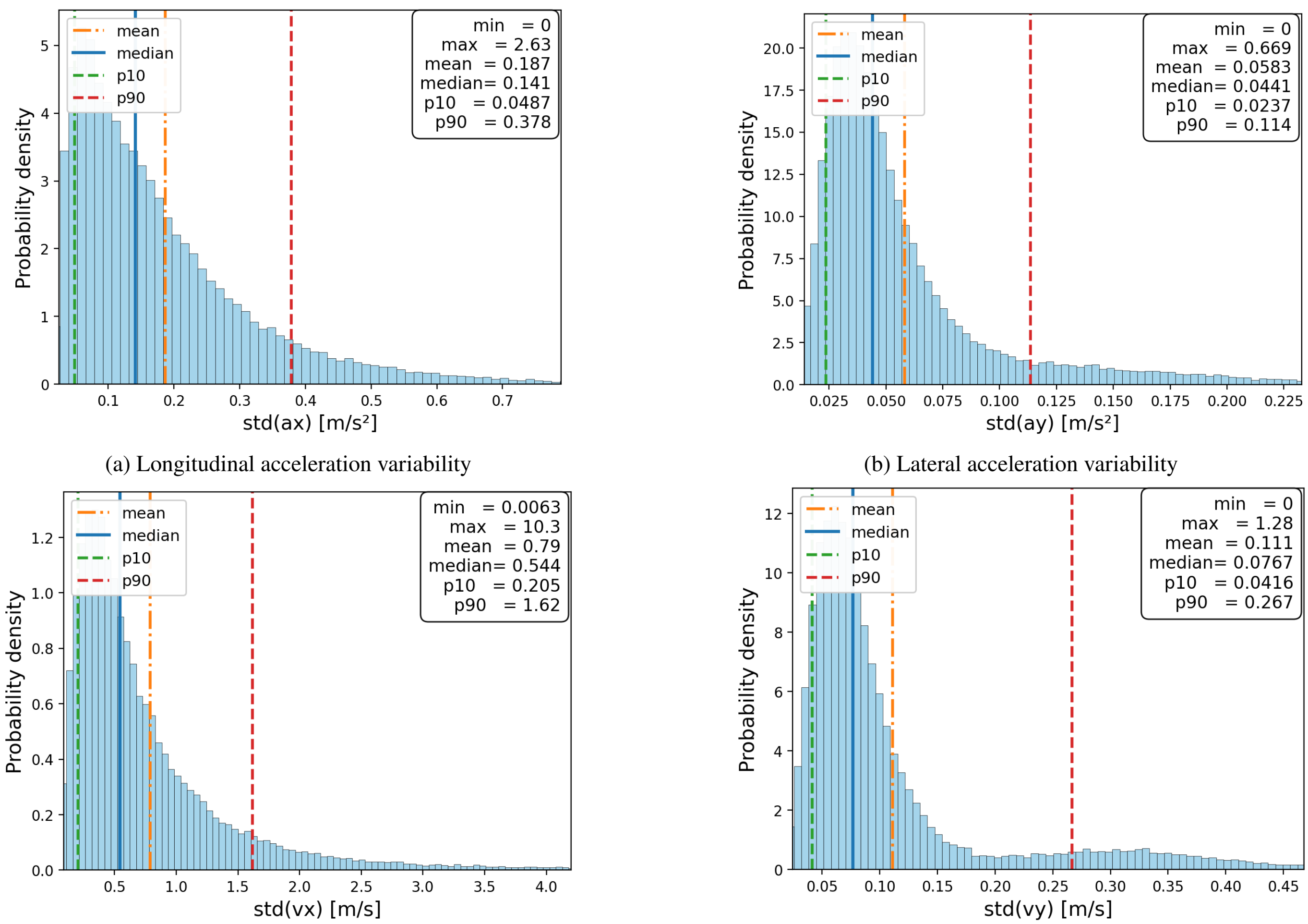
$$\begin{aligned} &p(S_{t+1} = s', A_{t+1} = a' \mid O_{1:t}) \\ &= \sum_{s,a} p(S_t = s, A_t = a \mid O_{1:t})\, p(S_{t+1} = s' \mid S_t = s) \\ &\quad \times p(A_{t+1} = a' \mid A_t = a, S_{t+1} = s'). \end{aligned} \tag{5}$$

The predicted maneuver is then obtained from the most likely joint state and its semantic maneuver label.

4. EXPERIMENTAL SETUP

4.1. Data Split

The dataset is partitioned at the vehicle level into disjoint training, validation, and test sets to prevent leakage across overlapping windows from the same trajectory. The split contains



(a) Longitudinal acceleration variability

(b) Lateral acceleration variability

(c) Longitudinal velocity variability

(d) Lateral velocity variability

FIGURE 4: Trajectory-level variability distributions of the main kinematic variables.

35,318 training vehicles (70%), 5,045 validation vehicles (10%), and 10,092 test vehicles (20%). The training set is used for parameter estimation, the validation set is used for early stopping, and the test set is reserved for final evaluation.

4.2. Evaluation Protocol

Evaluation is performed in two parts. First, held-out model fit is measured on complete unseen test sequences using likelihood-based metrics. This evaluates how well the trained DBN explains the observed window sequences. Second, online predictive evaluation is performed in a causal setting, where only observations up to the current time step are used to update the filtered belief and predict the next maneuver.

Because the DBN is trained without semantic maneuver labels, the learned joint latent states are interpreted after training through a two-stage procedure. First, for each joint state (s, a), posterior-weighted feature statistics are computed from the original window-level features in the training set. This involves computing the posterior probability that each training window belongs to that joint state, then computing weighted means and standard deviations of the features, giving a profile of the typical behavior associated with that state. Second, each joint state is assigned a semantic maneuver label by applying domain-based rules to its feature profile. These rules use feature thresholds and decision logic based on the patterns of longitudinal motion, lateral motion, front-vehicle presence, and interaction context to classify windows into one of eight maneuver classes.

Since the highD dataset does not provide maneuver labels directly, reference labels for online evaluation are generated from the preprocessed window features using a separate rule-based labeling procedure. These rules use cues such as longitudinal acceleration, lateral motion, lane-change evidence, and following behavior to assign each scored window to a maneuver class. The reference labels are used only for evaluation and are not used during EM training.

During online evaluation, the predicted label is obtained from the most likely DBN joint state through the corresponding semantic map and is compared with the rule-based reference label. Windows without a reliable reference label are excluded from scoring. These unclassified windows result from the rule set's limited coverage: they represent edge cases and boundary conditions that fall between the defined rule thresholds and thus cannot be reliably assigned to any single semantic class. The first five window observations of each test trajectory are used as

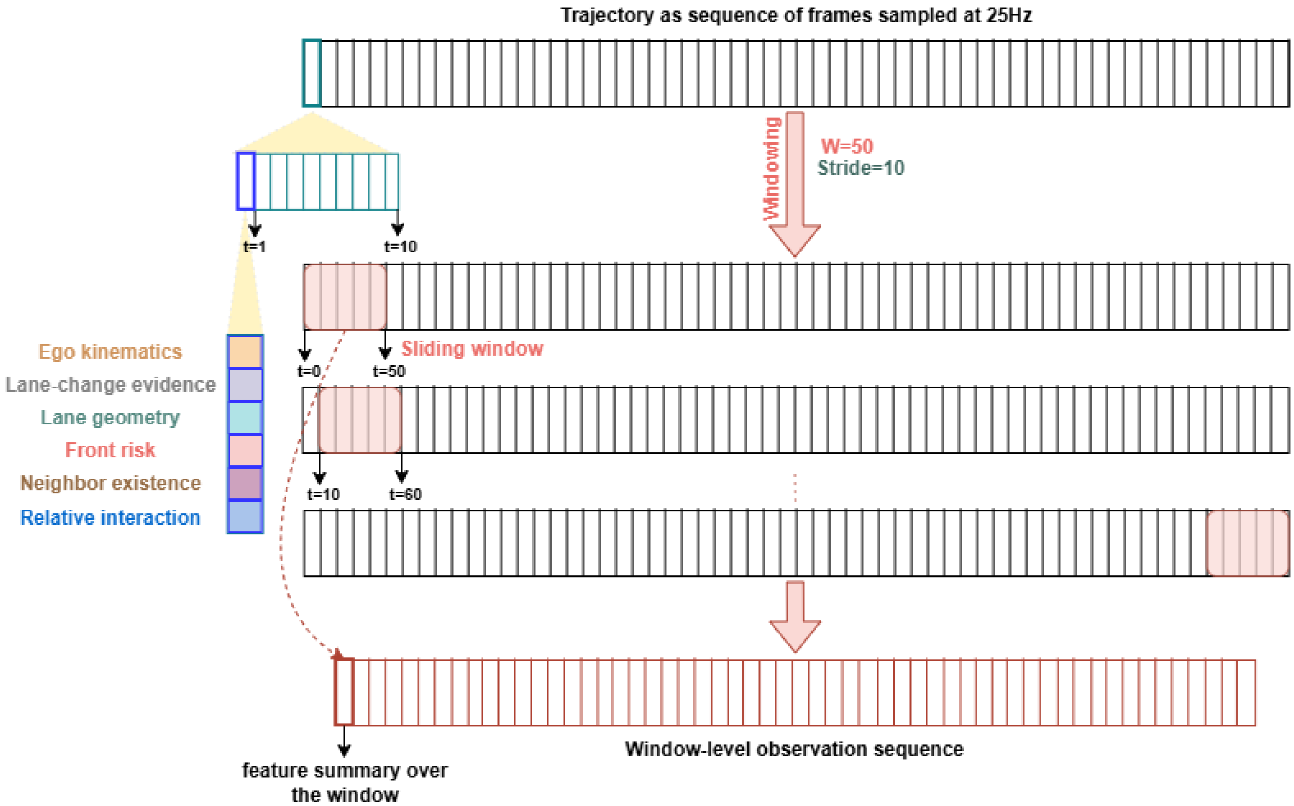


FIGURE 5: Illustration of the sliding-window representation

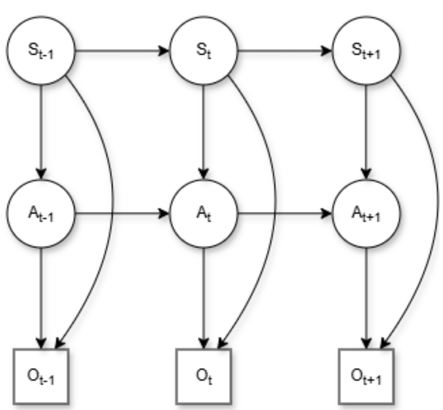


FIGURE 6: Structure of the proposed DBN

a warmup period and are also excluded. After warmup, one-step-ahead predictions are generated sequentially. For horizon-based evaluation, the forecast horizon covers the next five window steps, corresponding to 2.0 s.

4.3. Evaluation Metrics

Model fit on unseen test sequences is assessed using the average negative log-likelihood (NLL) per time step. NLL measures how much probability the trained DBN assigns to the observed window sequences in the held-out test set. It is therefore used as a likelihood-based sanity check for the probabilistic model: a lower NLL indicates better held-out likelihood and helps verify that the model has learned the main structure of the data without only fitting the training set. However, NLL is not used as the main evidence for safe maneuver prediction. Because naturalistic highway data are dominated by common behaviors such as cruising and lane keeping, a model can obtain a good NLL while still making poor predictions for rare but important events such as lane changes. For this reason, NLL is reported only as a model-fit metric and is interpreted together with predictive metrics, class-wise results, and horizon-based evaluation.

The Bayesian Information Criterion (BIC) is computed as:

$$\mathrm{BIC} = k \log N - 2 \log \hat{L}, \tag{6}$$

where k is the number of free model parameters, N is the total number of training time steps, and $\hat{L}$ is the fitted likelihood. BIC provides a complexity-penalized model-selection score; lower values indicate better trade-off between fit and complexity. Posterior occupancy diagnostics are used separately to inspect how effectively the learned latent states are used.

Online predictive performance is assessed using exact next-step accuracy, macro-averaged precision, recall, and F1-score. A horizon-based hit rate over 2.0 s is also reported. This metric counts a prediction as successful if the predicted semantic maneuver appears at least once within the next five window steps. It helps capture cases where a maneuver is predicted correctly but slightly shifted in time.

Average CPU latency is also measured for online belief update and next-step prediction. This is used to check whether the

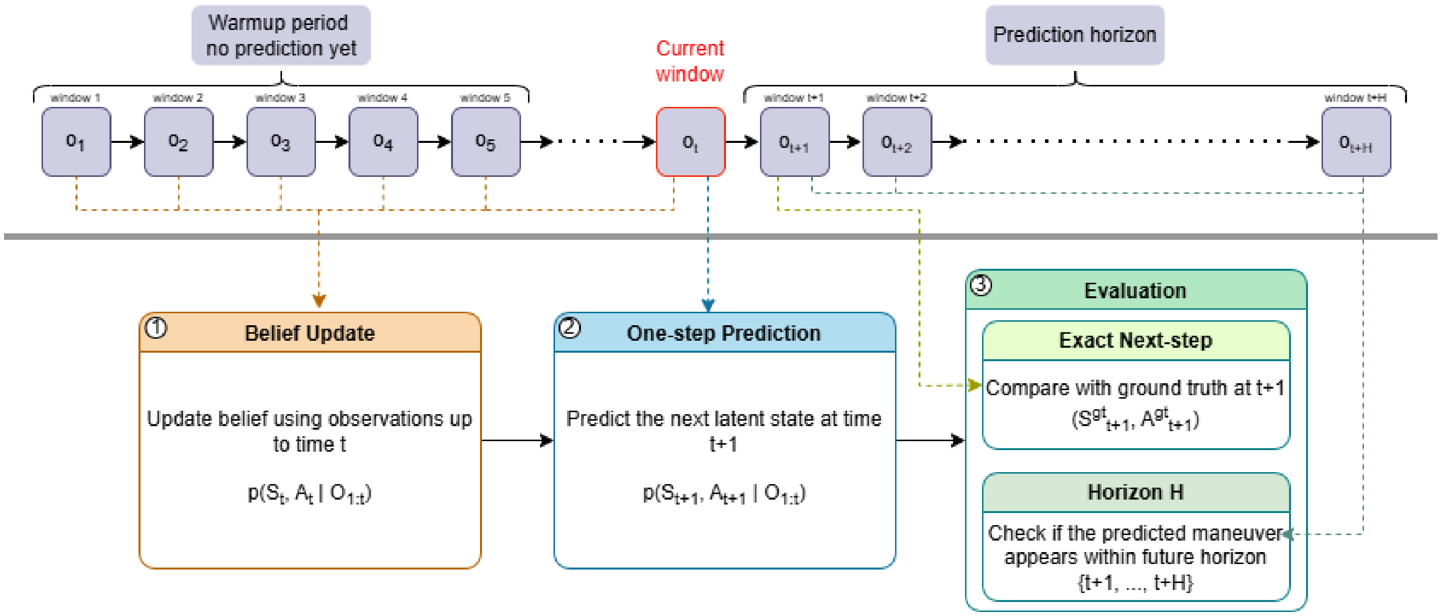


FIGURE 7: Causal online prediction pipeline used for one-step-ahead maneuver prediction

method can run faster than the 0.4 s interval between consecutive window observations.

5. RESULTS AND DISCUSSION

This section presents the results of the proposed DBN-based maneuver prediction framework.

5.1. Architecture Validation Study

To justify the selected model structure, an architecture validation study was carried out before the final evaluation. The main study compared 24 configurations with the same preprocessing pipeline, vehicle-level split, window representation, and EM training procedure. These configurations varied four design factors: (1) emission-model formulation, comparing Product-of-Experts (PoE) [35] with hierarchical emissions, (2) initial latent-state distribution, comparing fixed uniform and learned initial distributions, (3) lane-change weighting, comparing unweighted and lane-change-weighted variants, and (4) the inclusion of Bernoulli likelihood terms for binary observation features.

Across all 12 matched PoE–hierarchical comparisons, the hierarchical emission model achieved lower NLL than the corresponding PoE model. The same trend was observed for the BIC score, even though the hierarchical model uses more parameters. This indicates that the added flexibility of the hierarchical emission model is useful for the window-based highD observations, while the PoE formulation is too restrictive for the selected feature representation.

Lane-change events are rare compared with non-lane-change motion in the selected highD subset. Therefore, maximum-likelihood training can be dominated by the more frequent non-lane-change windows. To study this effect, lane-change weighting gives windows with lane-change evidence a higher weight in the statistics used for parameter updates. This gives rare lateral maneuver evidence more influence during training, while keeping the online inference and test evaluation unchanged.

Posterior diagnostics also supported the hierarchical model. In the unweighted hierarchical setting, the model used the joint latent-state space more evenly, with an effective joint-state occupancy of about 6.8 out of 8 possible states and a maximum joint-state occupancy of about 0.22. In comparison, several PoE and lane-change-weighted variants concentrated more probability mass on fewer joint states. This suggests that the hierarchical model gives a more balanced latent representation.

The remaining design factors had smaller or less consistent effects. Learning the initial distribution gave only minor changes, especially within the hierarchical family. Bernoulli likelihood terms improved the selected unweighted hierarchical variant, but they did not give a uniform gain across all configurations. Lane-change weighting did not improve the final model choice: the stronger weighting variant reduced the effective use of the latent-state space and gave worse model-fit scores.

A separate latent-cardinality ablation was then used to compare the selected S2A4 (2 style, 4 action) structure with 1S4A, 1S5A, 2S3A, and 2S5A variants. The one-style models performed worse, showing that the style variable is useful for separating broader interaction context from short-term maneuver dynamics. The 2S3A model was too coarse, while 2S5A improved likelihood but produced a more fine-grained and less compact semantic decomposition. Therefore, S2A4 was selected as the best trade-off among the tested structures.

Based on these validation results, the final model used in this paper is the unweighted hierarchical S2A4 DBN with a fixed uniform initial distribution, and Bernoulli likelihood terms enabled.

5.2. Semantic Labels

For the model, the two style states are interpreted after training as low-interaction and high-interaction driving regimes. The low-interaction style represents driving with weaker front-vehicle influence, while the high-interaction style represents more constrained driving with stronger coupling to nearby traffic. The action-state meanings used for evaluation are shown in Table 1.

TABLE 1: Action semantics.

Action state	Semantic meaning
a_0	Cruising
a_1	Braking
a_2	Acceleration with left lateral tendency
a_3	Acceleration with right lateral tendency

5.3. Model Fit and Latent-State Usage

Table 2 shows the held-out model fit and latent-state usage. Here, k_{eff}^{joint} denotes the effective number of occupied joint latent states, and $occ_{\max}^{joint}$ denotes the maximum posterior occupancy of any joint state.

TABLE 2: Held-out fit and latent-state usage.

NLL/step	BIC [M]	k_{eff}^{joint}	$occ_{\max}^{joint}$
23.88	50.82	6.82	0.22

The model achieves an NLL of 23.88 and a BIC of 50.82 M on the held-out test set. The latent state usage is well-balanced, with $k_{\text{eff}}^{joint} = 6.82$ effective states out of 8 possible joint states, and maximum occupancy $occ_{\max}^{joint} = 0.22$, indicating that no single state dominates. This balanced occupancy pattern suggests that the learned representation effectively captures multiple distinct behavioral modes without over-relying on a few states.

5.4. Online Prediction Performance

Table 3 summarizes the online prediction results.

TABLE 3: Online prediction performance.

Accuracy	Precision	Recall	F1	Hit rate
79.4%	67.6%	82.0%	71.5%	81.3%

The model gives strong prediction performance across all metrics. Exact next-step accuracy reaches 79.4%, with a macro-averaged F1-score of 71.5%, demonstrating reliable joint style-action prediction. The macro recall of 82.0% shows that the model recovers most classes well on average. The lower macro precision of 67.6% indicates that some classes, especially semantically close behaviors, are still confused with each other. For a driving application, predictions can remain useful even if the correct behavior appears slightly later. The 2 s horizon hit rate of 81.3% indicates that the model recovers the correct behavior within the next five windows for most test cases. The mean time-to-hit is 0.42 s, showing that when correct, the predicted behavior typically manifests almost immediately.

Figure 8 shows the confusion matrix for the model. The diagonal structure is clear, which means that the model predicts most joint states correctly. The strongest classes are high-interaction braking, high-interaction cruising, and the two high-interaction acceleration-with-lateral-tendency states.

TABLE 4: Class-wise precision and recall.

Class	Precision	Recall
LI/cruising	42.9%	95.7%
LI/braking	65.7%	79.9%
LI/accel. with left lateral tendency	60.3%	82.4%
LI/accel. with right lateral tendency	62.6%	73.7%
HI/cruising	40.1%	95.3%
HI/braking	99.7%	81.0%
HI/accel. with left lateral tendency	81.3%	79.7%
HI/accel. with right lateral tendency	88.3%	68.5%

Table 4 gives the class-wise precision and recall for the model. The two cruising classes have high recall (95.7% and 95.3%), meaning that most cruising samples are recovered correctly. Their precision values are lower (40.1-42.9%), which is reasonable because weak braking or early lateral motion can appear similar to cruising within a short observation window.

The high-interaction braking class gives the strongest precision, while both braking classes show strong recall. The two acceleration-with-lateral-tendency classes are also predicted well, although the high-interaction right-lateral-tendency class has lower recall.

5.5. Computational Efficiency

The online prediction step is computationally light. The input data are sampled at 25 Hz and a new window is generated every 10 frames. Therefore, the available prediction interval is 0.4 s, or 400 ms.

The latency benchmark was carried out on a CPU-only setup using an AMD Ryzen 9 7950X 16-Core processor. In the benchmark, the CPU runtime for the window-to-prediction step is about 0.19 ms for one vehicle. The end-to-end CPU runtime, including frame-to-window construction and prediction, is about 0.61 ms for one vehicle. For five vehicles processed in batched mode, the end-to-end CPU runtime is about 1.97 ms. All of these values are far below the 400 ms interval.

TABLE 5: Mean online latency for the prediction pipeline on CPU.

Scenario	Pipeline	Mean time
Single vehicle	window→predict	0.19 ms
Single vehicle	frame→window→predict	0.61 ms
Batched, 5 vehicles	frame→window→predict	1.97 ms

These results show that the proposed DBN can run in real time on standard CPU hardware. Even the end-to-end latency remains much smaller than the available 400 ms update interval.

5.6. Discussion

The proposed hierarchical DBN provides a well-balanced representation of human driving behavior. The architecture val-

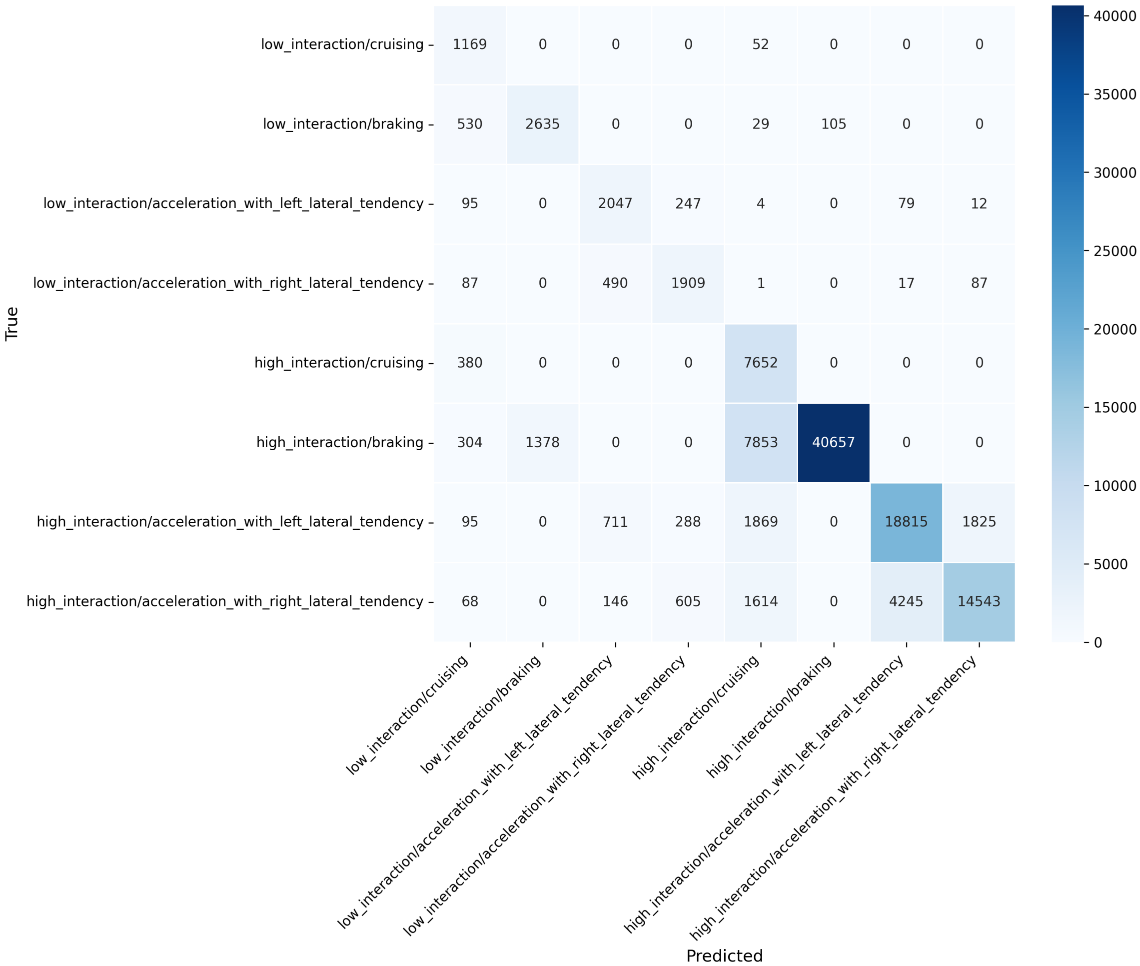


FIGURE 8: Confusion matrix.

idation study showed that the hierarchical S2A4 model gives the best trade-off among the tested configurations.

The model's balanced latent-state usage indicates that the learned representation captures multiple distinct behavioral modes without over-concentration on a single state. This balanced decomposition suggests that the S2A4 structure effectively separates driving regimes and maneuvers, supporting the interpretability of the learned latent space.

The model does have limitations. Some errors remain between semantically close behaviors, such as cruising and weak braking, which are difficult to distinguish within a short observation window. This suggests that the current observation design is better at capturing broader maneuver tendencies than very fine-grained behavior differences.

6. CONCLUSION

This paper presented a DBN-based framework for probabilistic short-horizon maneuver prediction of human-driven vehicles in highway traffic. The model uses window-based trajectory observations and predicts joint style-action states in a causal online setting. The style state represents the broader interaction regime, while the action state represents the current maneuver tendency.

The proposed hierarchical S2A4 DBN achieves a joint one-step accuracy of 79.4%, macro F1-score of 71.5%, and Hit rate of 81.3% over a 2 s forecast horizon. The model satisfies real-time requirements, with end-to-end CPU latency of 0.61 ms for single-vehicle prediction and 1.97 ms for five-vehicle batched prediction, both far below the 400 ms input interval. The architecture validation study further supports this model choice among the tested configurations.

Future work should test the framework on more diverse traffic datasets, improve the observation design for fine-grained maneuver separation, and study alternative latent-state structures. The probabilistic predictions can also be integrated into a larger automated-driving or AV decision-making pipeline to support risk assessment and safe interaction with human drivers.

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